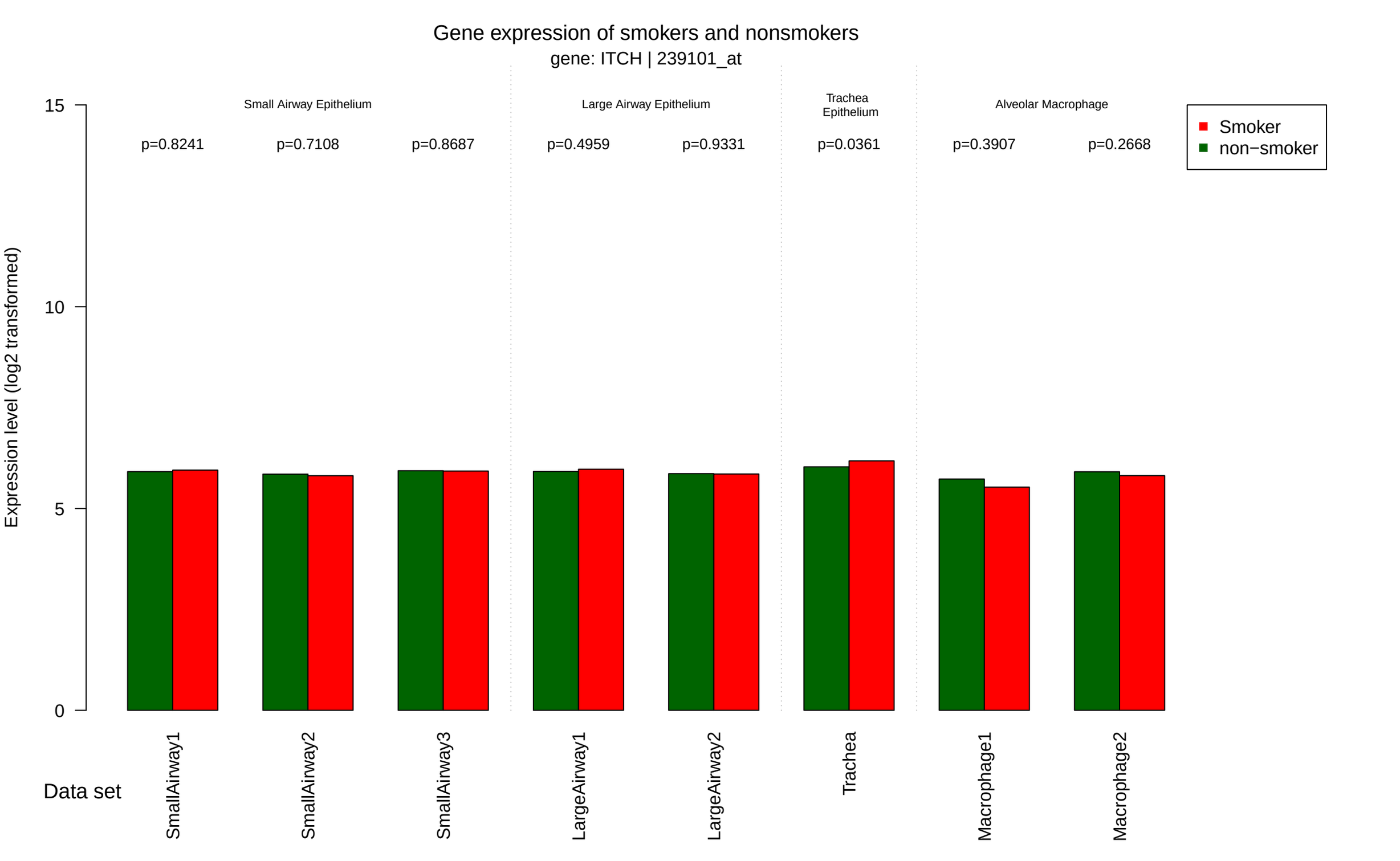
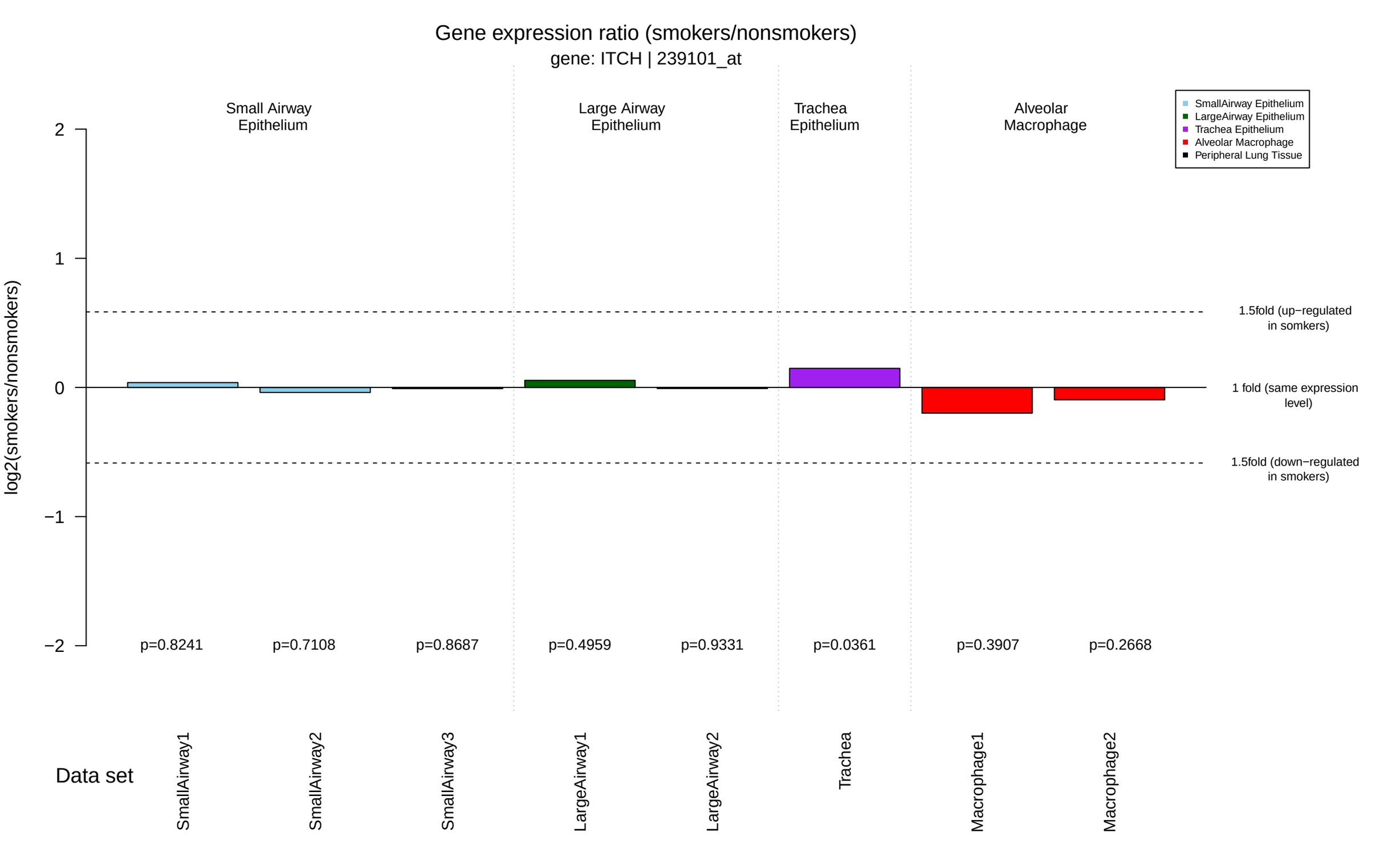
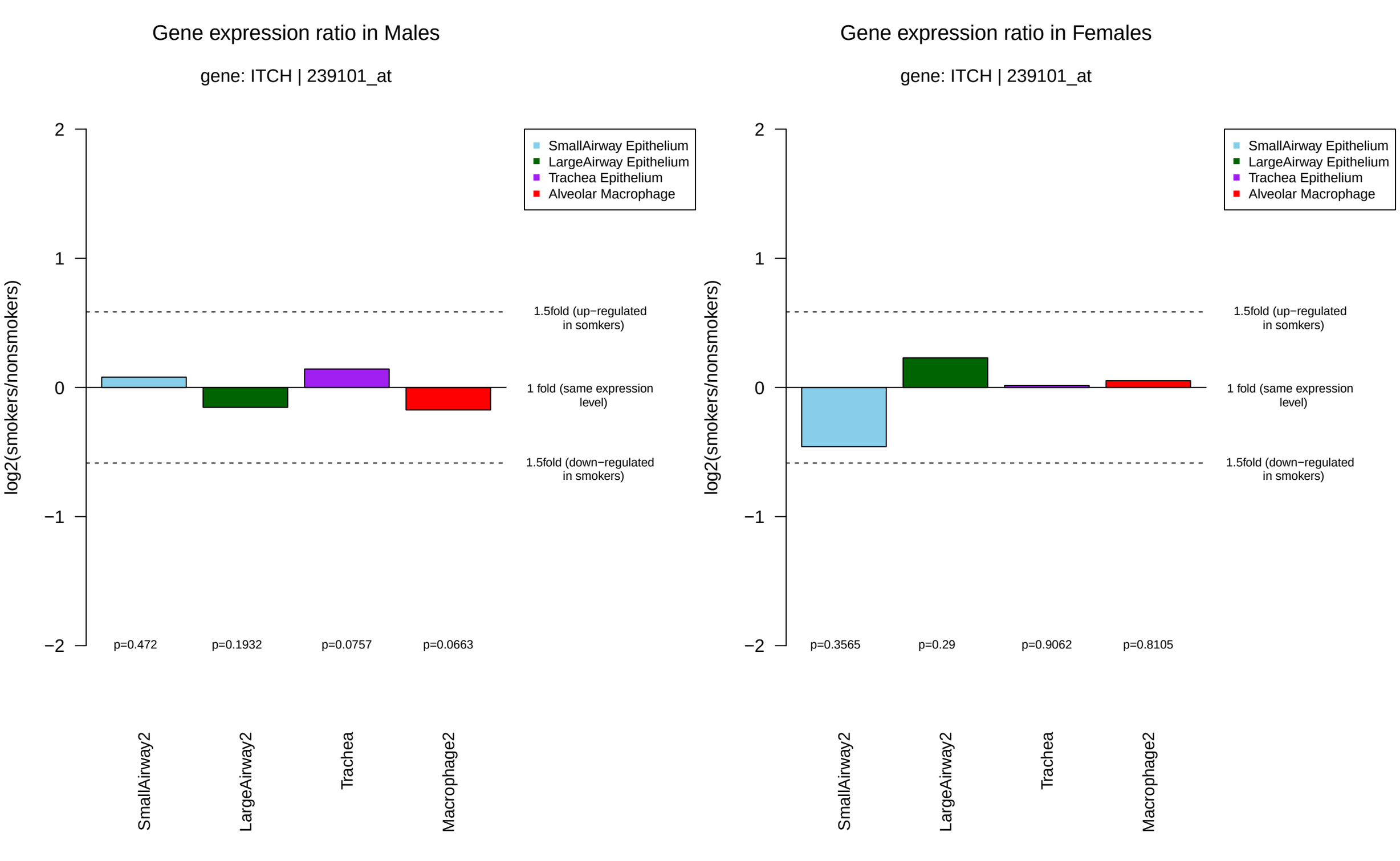
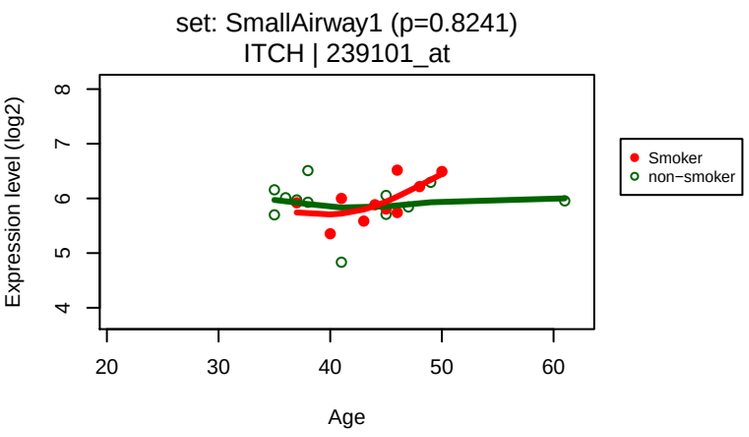
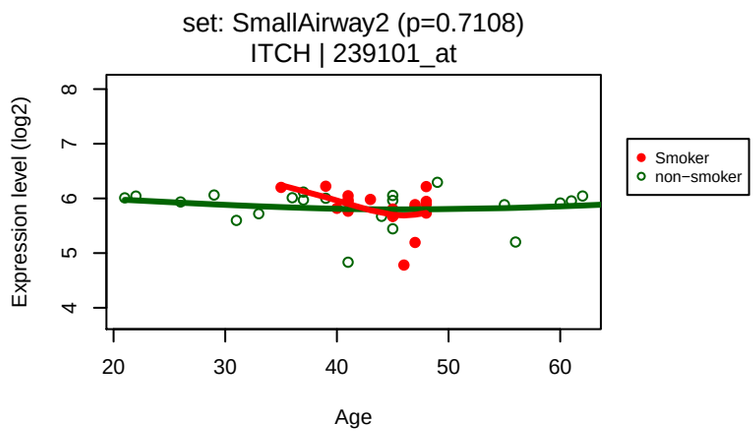
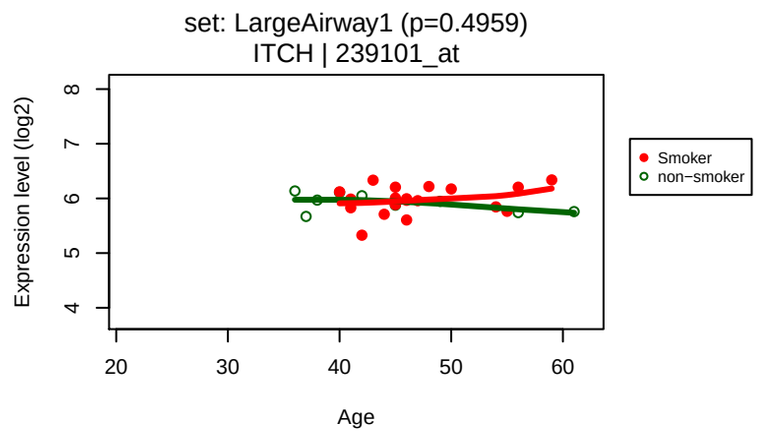

Gene expression of smokers and nonsmokers
gene: ITCH | 239101_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8241
p=0.7108
p=0.8687
p=0.4959
p=0.9331
p=0.0361
p=0.3907
p=0.2668
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

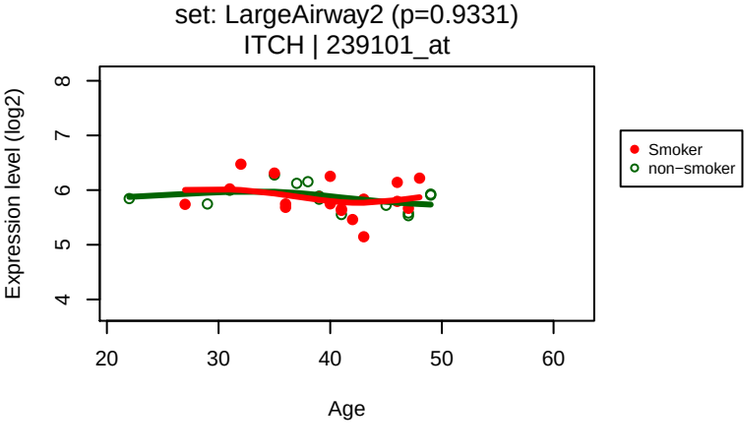

Gene expression ratio (smokers/nonsmokers)
gene: ITCH | 239101_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8241
p=0.7108
p=0.8687
p=0.4959
p=0.9331
p=0.0361
p=0.3907
p=0.2668
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

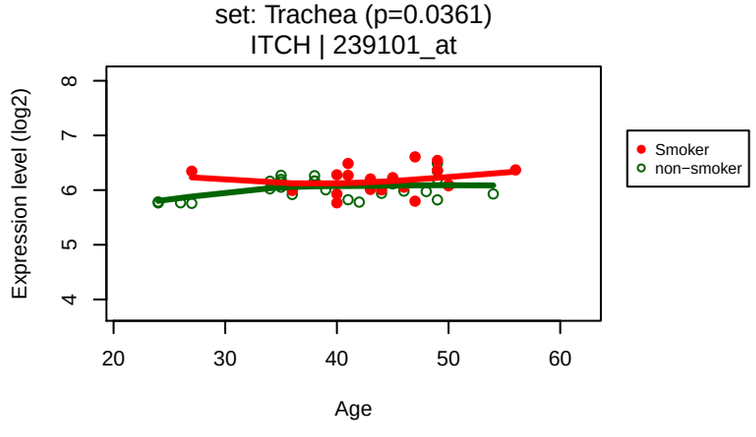

Gene expression ratio in Males
gene: ITCH | 239101_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.472
p=0.1932
p=0.0757
p=0.0663
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ITCH | 239101_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3565
p=0.29
p=0.9062
p=0.8105
SmallAirway2
LargeAirway2
Trachea
Macrophage2

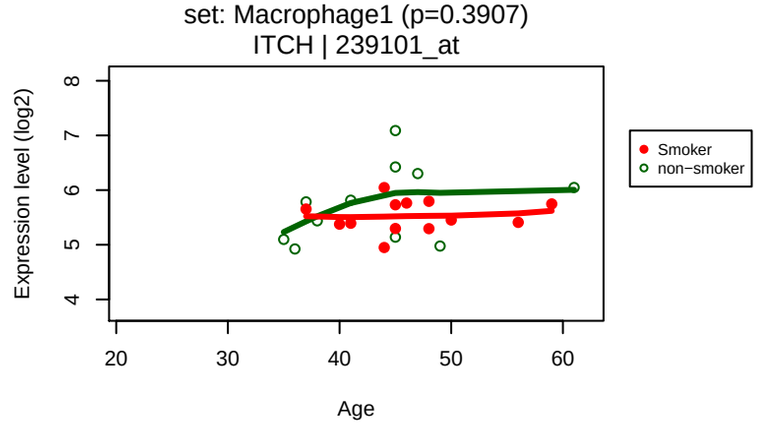

set: SmallAirway1 (p=0.8241)
ITCH | 239101_at
Expression level (log2)
Age
Smoker
non−smoker

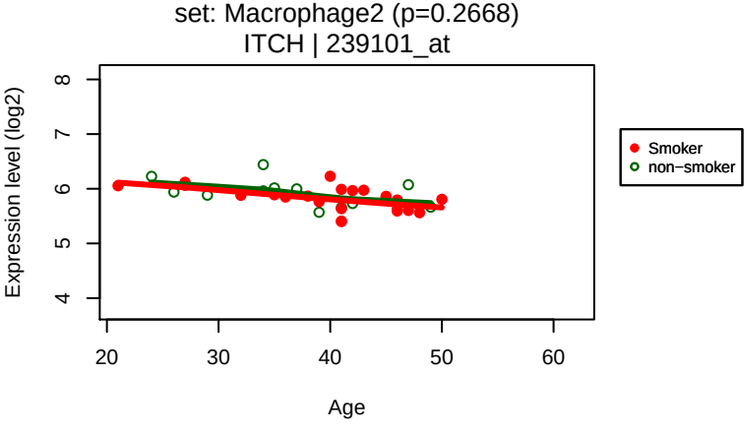

set: SmallAirway2 (p=0.7108)
ITCH | 239101_at
Expression level (log2)
Age
Smoker
non−smoker

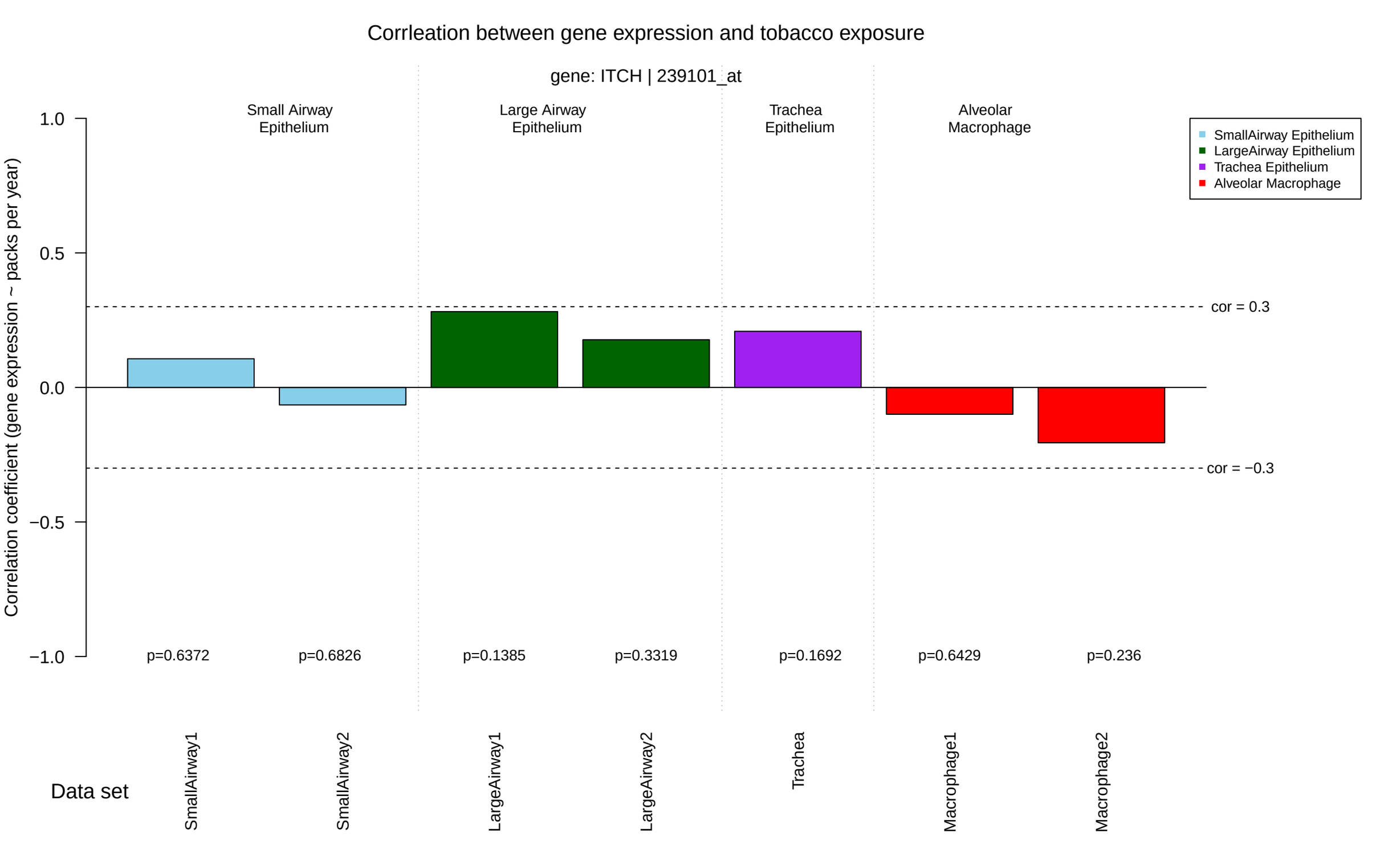

set: LargeAirway1 (p=0.4959)
ITCH | 239101_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9331)
ITCH | 239101_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0361)
ITCH | 239101_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3907)
ITCH | 239101_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2668)
ITCH | 239101_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ITCH | 239101_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6372
p=0.6826
p=0.1385
p=0.3319
p=0.1692
p=0.6429
p=0.236
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2